

MODULE 3 - Relational Database Design

- 3.1 Design Guidelines for Relational Schemas ✓
 - Functional dependencies ✓
- 3.2 First Normal form ✓
 - Second Normal form ✓
 - Third Normal form ✓
- 3.3 Decomposition using functional dependencies ✓
 - Boyce Codd Normal form ✓
 - Decomposition using Multivalued Dependencies ✓
 - Fourth Normal form ✓
- 3.4 Database design and Implementation process.

→ Domain (set of permissible value in particular column) is a set of atomic values.

A column is also called as Domain / Field / Arity or degree

→ Table (Relation) is a set of tuples / rows / entities or records

Properties of Relational table

- ① Cells contains atomic values
- ② Values in a Column are of same kind
- ③ Each row is unique
- ④ No. of two tables can have the same name in relational schema
- ⑤ Each column has a unique name
- ⑥ The sequence of rows is insignificant.
- ⑦ The sequence of columns is insignificant.

Purpose of Normalization

- Normalization may be simply defined as a refinement process which includes creating tables and establishing relationships b/w those tables according to the rules design both to protect data and make the database more flexible by eliminating two factors

① Redundancy

② Inconsistent Dependency

- Without normalization the DB system may be inaccurate, slow and inefficient, they might not produce the data we expect. A series of normal form tests that can be carried out on individual relational schemas so that relational DB can be normalized to any desired degree.

→ 1NF >> 2NF >> 3NF >> BCNF

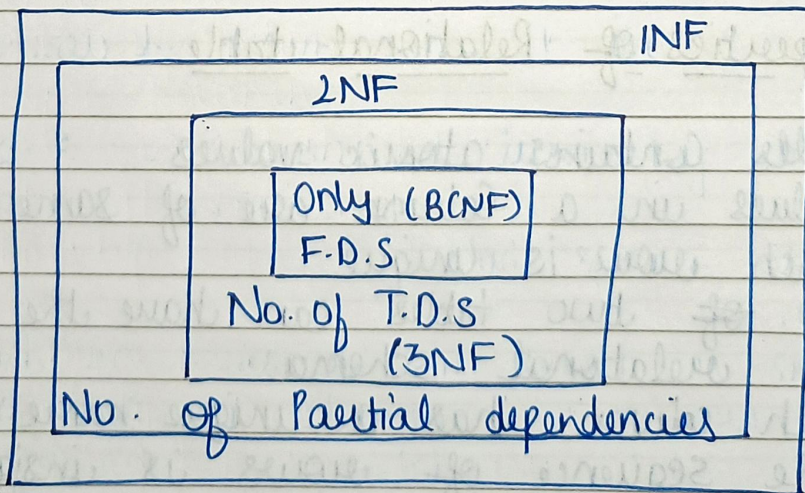


Fig:- Hierarchy of NF's

Functional Dependency

- A formal tool for analysis of relational schemas
- In a relation R , if ' α ' $\in R$ AND ' β ' $\in R$, then attributes on a set of attribute ' α ' functionally derives an attribute or set of attributes ' β ', if each ' α ' value is associated with precisely one ' β ' value
- For all pairs of tuples t_1 and t_2 in R s.t.
 - If $T_1[\alpha] = T_2[\alpha]$
then, $T_1[\beta] = T_2[\beta]$

eg]

X	Y	Z
1	4	2
1	4	3
2	6	3
3	2	2

X and Y have functional dependency.

→ Trivial functional dependency:- If β is a subset of α , then the functional dependency $\alpha \rightarrow \beta$ will always hold.

→ Transitive dependency:- A functional dependency for a non-prime attribute to non-prime attribute is called transitive.

$R(A, B, C, D)$ A is candidate Key

$A \rightarrow B$

$B \rightarrow C$ (transitive dependency)

$C \rightarrow D$ (transitive)

Teacher's Signature:.....

• Attribute Closure / Closure on Attribute Set

→ Attribute Closure of an attribute set can be defined as set of attributes which can be functionally determined from F.

• Denoted by F^+

eg) $R(A, B, C, D)$

$A \rightarrow B$

$B \rightarrow C$

$AB \rightarrow D$

$A^+ = \underline{\underline{ABCD}}$

$R(ABCDEFGH)$

$A \rightarrow B$

$BC \rightarrow DE$

$AEG \rightarrow H$

$(A)^+ = AC$

$= \underline{\underline{ABCDE}}$

FIRST NORMAL FORM

• 1NF is the initial step of DB normalization
Implications of first Normal form

• Atomic Value :- Each cell in a table contains indivisible, atomic values.

• Unique Columns :- Each column must have a distinct name to identify the data it contains

• Primary Key :- A table in 1NF should have a primary key that uniquely identifies each record.

• Eliminating Duplicates :- Duplicate rows are removed to prevent data redundancy.

- Partial Dependency:- When a non-prime attribute is dependent only on a part (Proper subset) of candidate key, then it is called partial dependency. (PRIME > NON-PRIME)
- Full Dependency:- When a non-prime attribute is dependent on the entire candidate key then it is called full dependency.

Second Normal form (2NF)

- Relation R is in 2NF if,
 - R should be in 1NF
 - R should not contain any Partial dependency. (That is every non-prime attribute should be fully dependent upon candidate Key)

eg) $R(A, B, C) \quad B \rightarrow C$

A	B	C	A	B	B	C
a	1	X	A	1	1	X
b	2	Y	B	2	2	Y
a	3	Z	A	3	3	Z
C	3	Z	C	3		
D	3	Z	D	3		
E	3	Z	E	3		

③ Third Normal Form (3NF)

Let R be a relational schema, it is said to be in 3NF

- R should be in 2NF
- It must not contain any transitive dependency
- A relational schema R is said to be in 3NF if every functional dependency in R from $\alpha \rightarrow \beta$, either α is super key or β is prime attribute.

eg:- $R(A, B, C)$

$A \rightarrow B$

$B \rightarrow C$

A	B	C		A	B	B	C
A	1	P		A	1	1	P
B	2	Q	}	B	2	2	Q
C	2	Q		C	2	3	R
D	2	Q	}	D	2	4	S
E	3	R		E	3		
F	3	R	}	F	3		
G	4	S		G	4		

→ BCNF (Boyce Codd Normal form)

A relation R is said to be BCNF if every functional dependency in R from $\alpha \rightarrow \beta$

- α must be a super key.

eg) $R(A, B, C)$

$AB \rightarrow C$

$C \rightarrow B$

A	B	C
A	C	B
B	B	C
B	A	D
A	A	E
C	C	B
D	C	B
E	C	B
F	C	B

A	B	C	B
A	B	B	C
B	B	C	B
B	A	D	A
A	A	E	A
C	C		
D	C		
E	C		
F	C		

- A relation with two attributes is always in BCNF.
- A relation schema R consists of only prime attributes then R is always in 3NF, but may or may not be in BCNF.

Q] $R(A, B, C, D, E, F, G, H, I, J)$

$F = AB \rightarrow C$

$A \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

$D \rightarrow IJ$

It is in 1NF as it contains partial dependency. To compose it into 2NF

$AB, ADE, BFGH, ADEIJ$

Multivalued Dependency

- Denoted by $A \twoheadrightarrow B$, Means, for every value of A , there may exist more than one value of B

eg:- $S_name \twoheadrightarrow Clubname$

Sname	Clubname
Ram	Dance
Ram	Guitar

→ Trivial Multivalued Dependency: - $X \twoheadrightarrow Y$ is one where either Y is a subset of X , or X and Y together form the whole set of attributes of the relation.

$S_name \twoheadrightarrow club_name$
 $S_name \twoheadrightarrow p_no$

S-name	Cname	p-no
Ram	Dance	123
Ram	Guitar	123
Ram	Singing	123 789
Ram	Coding	123 789

fig → non trivial dependency

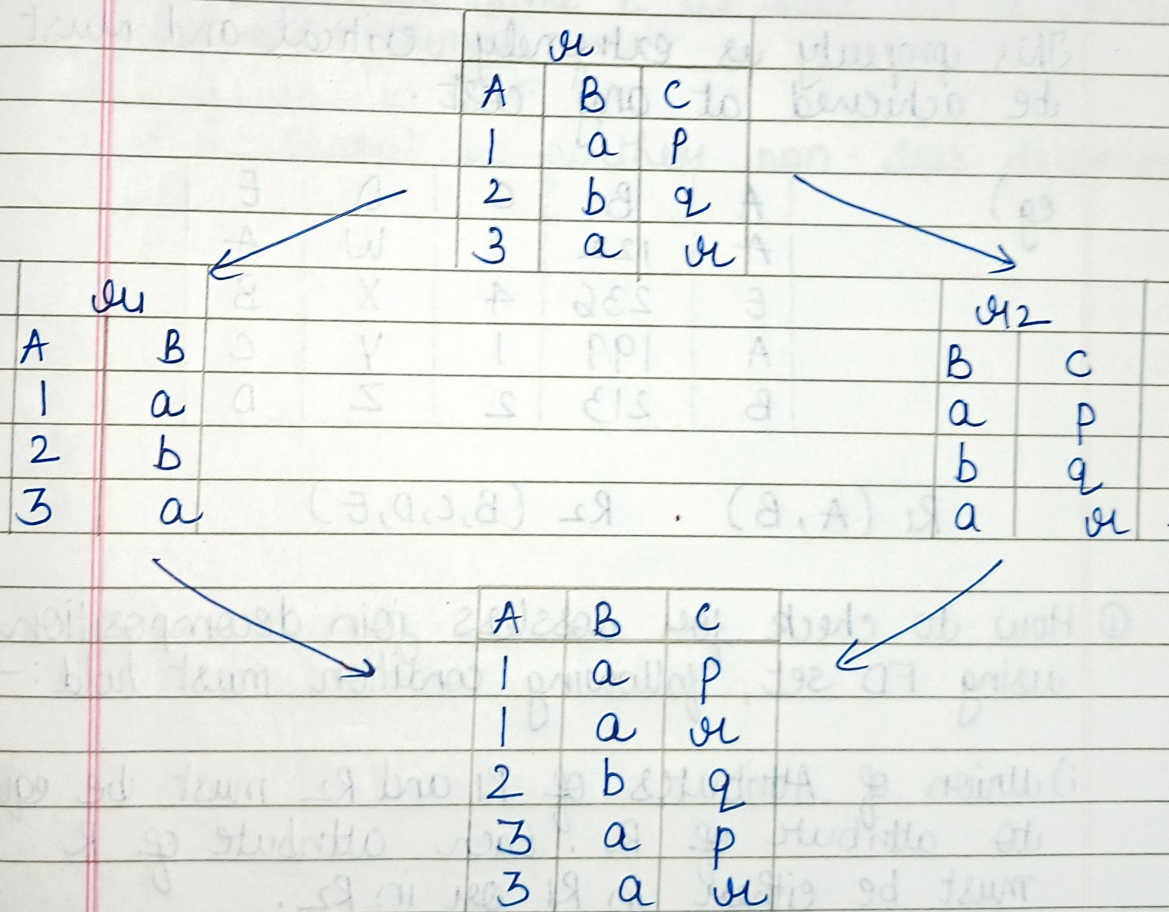
④ 4NF.

- It is in BCNF
- There must not exist any non-trivial multivalued dependency.

Each MVD is decomposed into separate tables, where it becomes trivial MVD.

Lossy / Lossless - Dependency

- Because of a normalization a table is decomposed into two or more tables, but during this decomposition we ensure satisfaction of some properties out of which the most imp is lossless join property / decomposition.



- Decomposition is lossless if $R_1 \bowtie R_2 = R$ "The decomposition of relation R into R_1 and R_2 is lossless when the join of R_1 and R_2 yield the same relation as in R ". which guarantees that the spurious tuple generation problem does not occur with respect to the relation schemas created after decomposition.

This property is extremely critical and must be achieved at any cost.

eg)

A	B	C	D	E
A	122	1	W	A
E	236	4	X	B
A	199	1	Y	C
B	213	2	Z	D

$R_1 (A, B)$

$R_2 (B, C, D, E)$

- How to check for lossless join decomposition using FD set, following condition must hold:-

i) Union of Attributes of R_1 and R_2 must be equal to attribute of R . Each attribute of R must be either in R_1 or in R_2 .

$$\text{Att}(R_1) \cup \text{Att}(R_2) = \text{Att}(R)$$

ii) Intersection of Attributes of R_1 and R_2 must not be NULL. $\text{Att}(R_1) \cap \text{Att}(R_2) \neq \emptyset$

iii) Common attribute must be a key for at least one relation (R_1 or R_2)

$$\text{Att}(R_1) \cap \text{Att}(R_2) \rightarrow (R_1)$$

$$\text{Att}(R_1) \cap \text{Att}(R_2) \rightarrow (R_2)$$

5NF / Project - Joint Normal form

- A Relational table R is said to be in 5th normal form if
 - it is in 4NF
 - it cannot be further non-loss decomposed.